

1. Which commands will be use to complete the given task with the help of the following operation?

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Terminating Process via Terminal

To reduce system load, Adam can use these specific commands:

* kill a single process by PID

- command: `kill <PID>`

- Example: `kill 1234`

* kill a process by name

If Adam knows the name but not the ID, he can use `pkill`

- command: `pkill <process-name>`

- Example: `pkill chrome`

* kill processes by name

The `killall` command is used to kill all instances of a process running with a specific name

- command: `killall <process-name>`

2. Write a program for process creation using C.

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1. Zombie Process :-

Create when the child process terminates but the parent process does not call `wait()` the child remain in the

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zombie state until the parent exist

- nano zombie.c → To create the zombie process program
- gcc zombie.c → to zombie + compile zombie program
- ps -el | grep → To check the zombie process status

2. Orphan Process:-

Creates when parent process terminates before the child process the child process continue execution and it adopted by the system

- nano orphan.c → To create and write a c program for orphan process.
- gcc orphan.c → -o orphan → To compile the program
- ./orphan → To execute the compiled program.

3. Create the process using fork() system call

- child process creation
- Parent process creation
- PPID and PID

- Process creating using fork()

Return Value of fork()

Meaning

1. Negative value creation of child process was unsuccessful

Zero (0)

Returned to the newly created child process

Positive value

Returned to the parent process

* PID (Process ID):-

A unique identifier for a specific process

* PPID (Parent Process ID):-

the ID of the process that created it.

Commands:

getpid(): Returns the PID of the current process

getppid(): Returns the PID of the parent process

Command →

Purpose.

- Ps - ef → Lists all running process with their PID and PPID
- pstree → Show processes in a visual tree format
- top → provides a real time view of union process
- kill -9 <PID> → forcefully terminates a process using it's PID.